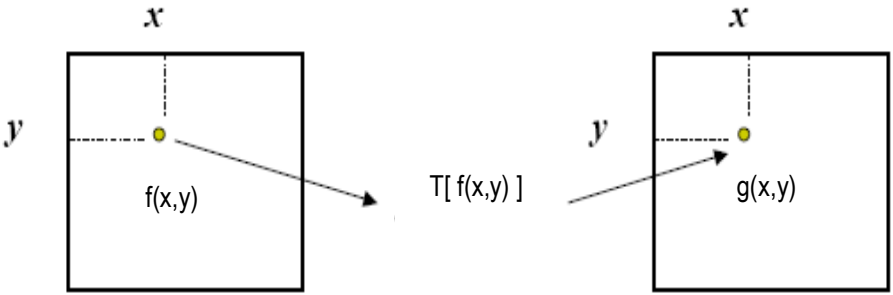
Procesamiento de imágenes médicas

Mejora de contraste mediante operaciones puntuales

Alejandro Veloz

Operadores puntuales

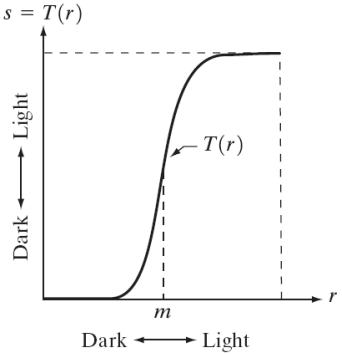


$$g(x,y) = T[\, f(x,y) \,]$$

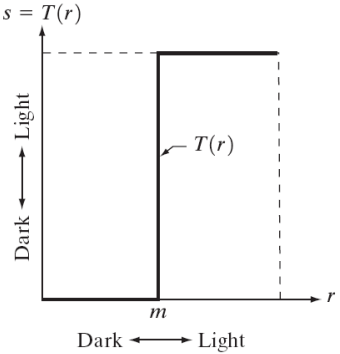
El valor de la **imagen de salida** g en el punto (x,y) sólo depende del valor de la **imagen de entrada** f en la posición (x,y).

Operadores puntuales

contrast stretching



Caso límite: *thresholding* function

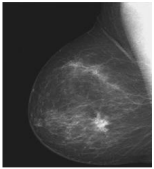


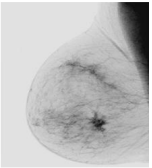
Operadores puntuales

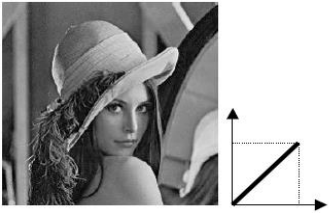
Operadores lineales:

(i) Operador identidad: $s = r$

(ii) Operador inverso o negativo: $s = L - 1 - r$





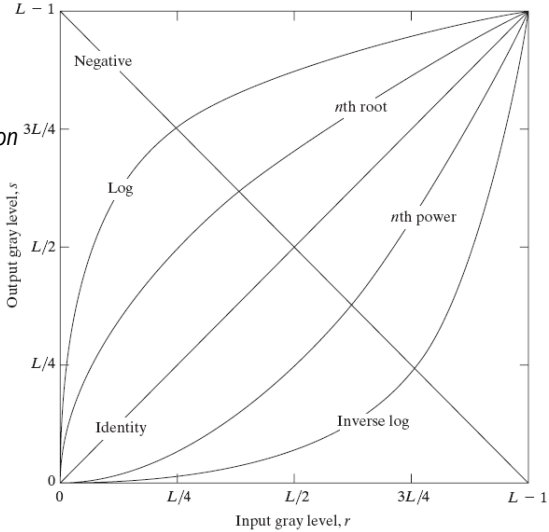


Operadores puntuales

Log-transformation

s = c log(1+r),

r ≥ 0



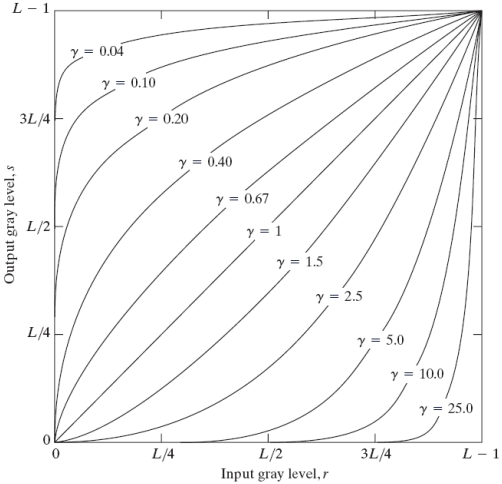
Operadores puntuales

Power-law transformation

$s = c\, r^{\gamma}$,

$c, \gamma > 0$

$s = c\,(r + \varepsilon)^{\gamma}$



Operadores puntuales

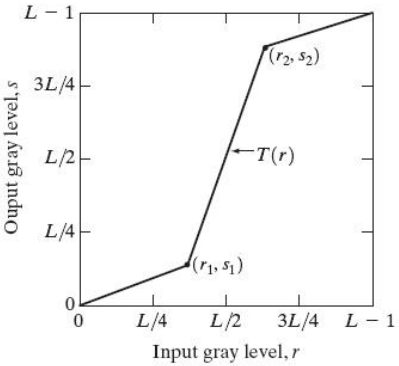
Power-law transformation function

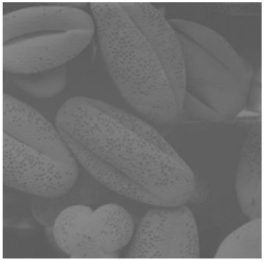
a b
c d

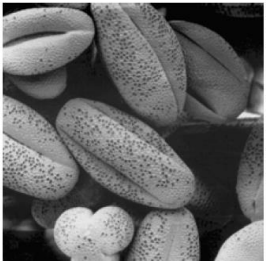
(a) Aerial image. (b)–(d) Results of applying the transformation in Eq. (3.2-3) with $c = 1$ and $\gamma = 3.0, 4.0$, and 5.0, respectively. (Original image for this example courtesy of NASA.)

Operadores puntuales

Piecewise-Linear Transformation Functions







Operadores puntuales

Piecewise-Linear Transformation Functions

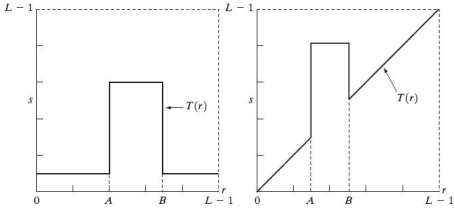
Gray-level slicing.

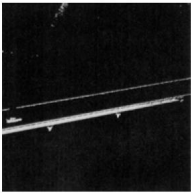
g(x,y) = k

para A ≤ f(x,y) ≤ B, con k, A y B en [0,L–1].

g(x,y) = f(x,y)

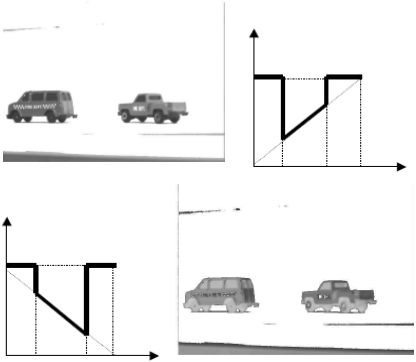
en otro caso.





Operadores puntuales

Piecewise-Linear Transformation Functions



Operadores puntuales

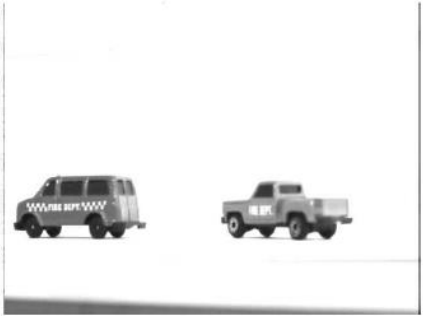
Piecewise-Linear Transformation Functions

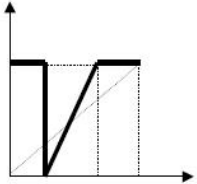
$g(x,y) = (L - 1)\,(f(x,y) - A) / (B - A)$

para $A \leq f(x,y) \leq B$, con A y B en $[0,L-1]$.

$g(x,y) = L - 1$

en otro caso.





Operadores puntuales

Piecewise-Linear Transformation Functions

Reducción de niveles de gris.

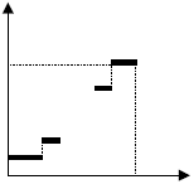
$g(x,y) = k_1$ para $0 \leq f(x,y) \leq th_1$,

$g(x,y) = k_2$ para $th_1 < f(x,y) \leq th_2$,

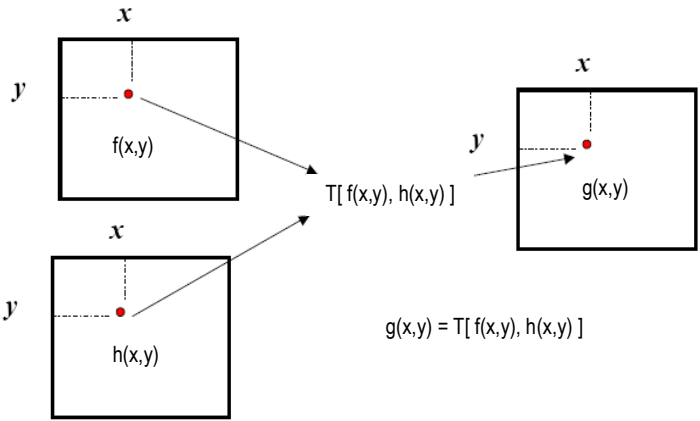
…

$g(x,y) = k_n$ para $th_n < f(x,y) \leq L - 1$,

con th_1 , th_2 , …, th_n en $[0, L-1]$



Operadores puntuales



$$g(x,y) = T[\, f(x,y), h(x,y) \,]$$

Operadores puntuales

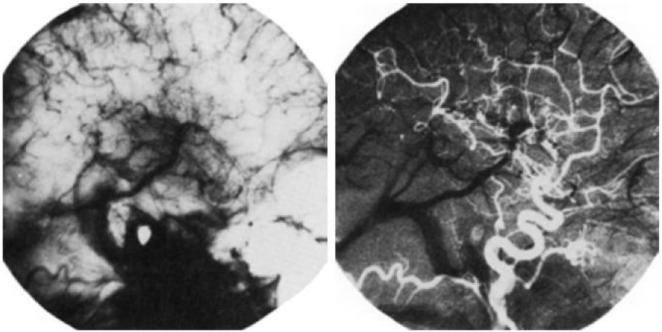
$$g(x,y) = [\, f(x,y) + h(x,y) \,] / 2$$

Operadores puntuales

$$g(x,y) = f(x,y) - h(x,y)$$

Mask mode radiography

a b



Enhancement by image subtraction. (a) Mask image. (b) An image (taken after injection of a contrast medium into the bloodstream) with mask subtracted out.